Type MS · Log equations: combine into one log, then remove the log

Answer 1 of 16 · Question 10 part whole · 5 marks

2022 JUN · Paper 02 · find part whole in this mark scheme extract

The mark scheme extract for this part is not available yet. Use "Check my answer" on the website.

Type MS · Log equations: combine into one log, then remove the log

Answer 2 of 16 · Question 8 part a · 3 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Question number		
8 (a)	$\log_a n = \log_a 3(2n-1)$	M1
	$n = 3(2n-1)$	M1
	$n = \frac{3}{5}$	A1 (3)
(b)(i)	$x = p^3$	B1 (1)
(b)(ii)	$\log_p y - \log_p 2^3 = 4 \Rightarrow \log_p\left(\frac{y}{2^3}\right) = 4$ or $\log_p\left(\frac{y}{8}\right) = 4$	M1
	$\frac{y}{2^3} = p^4 \Rightarrow \left(y = 2^3 p^4 \text{ or } 8p^4\right)$	M1
	$xy = 8p^7$	M1A1 (4)
	ALT (b)(ii)	
	$\log_p x + \log_p y - 3\log_p 2 = 4 + 3 \Rightarrow \log_p\left(\frac{xy}{2^3}\right) = 7$	{M1}
	$\frac{xy}{2^3} = p^7$	{M1}
	$xy = 8p^7$	{M1A1} (4)
		Total 8 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 2 of 16 · Question 8 part a · 3 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	M1	Uses the addition law of logs correctly $\log_a n = \log_a 3 + \log_a (2n-1) \Rightarrow \log_a n = \log_a 3(2n-1)$ Accept also $\log_a n = \log_a 3 + \log_a (2n-1) \Rightarrow 0 = \log_a \left(\frac{n}{3(2n-1)}\right) = (\log_a 1)$
	M1	For obtaining a linear equation from their log equation and attempting to find a value for n. $n = 3(2n-1)$ leading to a numerical value for n
	A1	For $n = \frac{3}{5}$
(b)(i)	B1	For $x = p^3$
(b)(ii)	M1	For stating that $3\log_p 2 = \log_p 8$ or $\log_p 2^3$ and for using the addition law correctly to combine the LHS: $\log_p y - \log_p 2^3 = 4 \Rightarrow \log_p \left(\frac{y}{2^3}\right) = 4$ or $\log_p \left(\frac{y}{8}\right) = 4$
	M1	Correctly removes logs on both sides to obtain: $\frac{y}{2^3} = p^4 \Rightarrow \left(y = 2^3 p^4 \text{ or } 8p^4\right)$
	M1	For correctly finding the product of their x and their y: $xy = 'p^3' \times '8p^4'$
	A1	For the correct answer of $xy = 8p^7$
	ALT	
	M1	For stating that $3\log_p 2 = \log_p 8$ or $\log_p 2^3$ and states $\log_p x + \log_p y - 3\log_p 2 = 3 + 4$ Uses the addition law correctly to combine the LHS $\log_p x + \log_p y - 3\log_p 2 = 4 + 3 \Rightarrow \log_p \left(\frac{xy}{2^3}\right) = 7$
	M1	Correctly remove logs on both sides to obtain: $\frac{xy}{2^3} = p^7$
	M1	Correctly rearrange their expression to make xy the subject
	A1	For the correct answer of $xy = 8p^7$

Type MS · Log equations: combine into one log, then remove the log

Answer 2 of 16 · Question 8 part a · 3 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 3 of 16 · Question 8 part b.ii · 4 marks

2021 JUN · Paper 01 · extract 1 of 3 · find part b.ii in this mark scheme extract

8 (a)	$\log_a n = \log_a 3(2n-1)$	M1
	$n = 3(2n-1)$	M1
	$n = \frac{3}{5}$	A1 (3)
(b)(i)	$x = p^3$	B1 (1)
(b)(ii)	$\log_p y - \log_p 2^3 = 4 \Rightarrow \log_p\left(\frac{y}{2^3}\right) = 4$ or $\log_p\left(\frac{y}{8}\right) = 4$	M1
	$\frac{y}{2^3} = p^4 \Rightarrow \left(y = 2^3 p^4 \text{ or } 8p^4\right)$	M1
	$xy = 8p^7$	M1A1 (4)
	ALT (b)(ii)	
	$\log_p x + \log_p y - 3\log_p 2 = 4 + 3 \Rightarrow \log_p\left(\frac{xy}{2^3}\right) = 7$	{M1}
	$\frac{xy}{2^3} = p^7$	{M1}
	$xy = 8p^7$	{M1A1} (4)
		Total 8 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 3 of 16 · Question 8 part b.ii · 4 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part b.ii in this mark scheme extract

Part	Mark	Notes
(a)	M1	Uses the addition law of logs correctly $\log_a n = \log_a 3 + \log_a (2n-1) \Rightarrow \log_a n = \log_a 3(2n-1)$ Accept also $\log_a n = \log_a 3 + \log_a (2n-1) \Rightarrow 0 = \log_a \left(\frac{n}{3(2n-1)}\right) = (\log_a 1)$
	M1	For obtaining a linear equation from their log equation and attempting to find a value for n. $n = 3(2n-1)$ leading to a numerical value for n
	A1	For $n = \frac{3}{5}$
(b)(i)	B1	For $x = p^3$
(b)(ii)	M1	For stating that $3\log_p 2 = \log_p 8$ or $\log_p 2^3$ and for using the addition law correctly to combine the LHS: $\log_p y - \log_p 2^3 = 4 \Rightarrow \log_p \left(\frac{y}{2^3}\right) = 4$ or $\log_p \left(\frac{y}{8}\right) = 4$
	M1	Correctly removes logs on both sides to obtain: $\frac{y}{2^3} = p^4 \Rightarrow \left(y = 2^3 p^4 \text{ or } 8p^4\right)$
	M1	For correctly finding the product of their x and their y: $xy = '\,p^3\,' \times '\,8p^4\,'$
	A1	For the correct answer of $xy = 8p^7$
	ALT	
	M1	For stating that $3\log_p 2 = \log_p 8$ or $\log_p 2^3$ and states $\log_p x + \log_p y - 3\log_p 2 = 3 + 4$ Uses the addition law correctly to combine the LHS $\log_p x + \log_p y - 3\log_p 2 = 4 + 3 \Rightarrow \log_p \left(\frac{xy}{2^3}\right) = 7$
	M1	Correctly remove logs on both sides to obtain: $\frac{xy}{2^3} = p^7$
	M1	Correctly rearrange their expression to make xy the subject
	A1	For the correct answer of $xy = 8p^7$

Type MS · Log equations: combine into one log, then remove the log

Answer 3 of 16 · Question 8 part b.ii · 4 marks

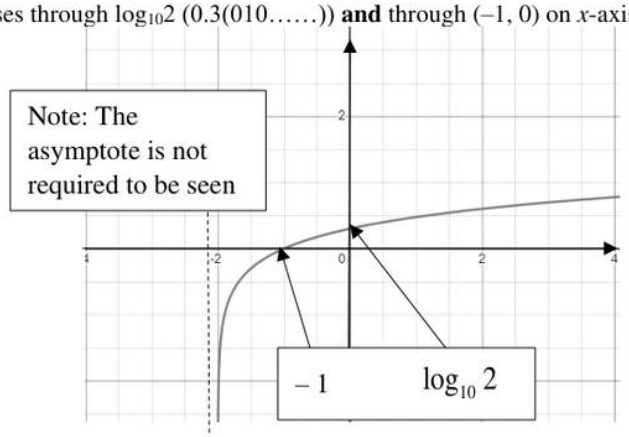
2021 JUN · Paper 01 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 4 of 16 · Question 7 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 1 of 3 · find part b in this mark scheme extract

7 (a)	Shape of the curve, **crossing** negative x-axis and positive y-axis with the asymptotic nature of the curve shown. passes through $\log_{10}2$ (0.3(010……)) **and** through (–1, 0) on x-axis  	B1 B1 [2]
(b)	$\left(2\log_a 4+2\log_a 4^2\right)$ or $(2\log_a 2^2+2\log_a 2^4)$ $2\log_a 4+4\log_a 4$ or $4\log_a 2+8\log_a 2$ $\log_a 4=\frac{1}{6}$ or $\log_a 2=\frac{1}{12}$ oe $a^{\frac{1}{6}}=4$ or $a^{\frac{1}{12}}=2$ $a=4096$ **ALT** $(2)\log_a 64=1\Rightarrow\log_a 64=\frac{1}{2}$ $\Rightarrow a^{\frac{1}{2}}=64$ $\Rightarrow a=4096$	M1 dM1 A1 [M1 dM1 A1] [3]
(c)	$\log_q 16\rightarrow\frac{\log_2 16}{\log_2 q}$ $\left(5\frac{\log_2 16}{\log_2 q}+4\log_2 q=24\right)$ $20+4(\log_2 q)^2=24\log_2 q$ $(\log_2 q-5)(\log_2 q-1)(=0)$ $(\log_2 q=5)\rightarrow q=32$ $(\log_2 q=1)\rightarrow q=2$	$\log_2 q\rightarrow\frac{\log_q q}{\log_q 2}$ or $\frac{1}{\log_q 2}$ $\left(5\log_q 16+4\frac{\log_q q}{\log_q 2}=24\right)$ $20(\log_q 2)^2+4=24\log_q 2$ $(5\log_q 2-1)(\log_q 2-1)(=0)$ $\left(\log_q 2=\frac{1}{5}\right)\rightarrow q=32$ $(\log_q 2=1)\rightarrow q=2$	M1 M1 A1 dM1 M1 A1 [6]
		Total 11 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 4 of 16 · Question 7 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Additional Guidance
(a)	B1	The curve must not bend back on itself and should cross both axes once on negative x and once on positive y. It needs to demonstrate an asymptotic nature. Accept an asymptote drawn correctly as well as their curve.
	B1	It is enough for -1 to be indicated on the x-axis with the curve passing through this point and for $\log_{10}2$ to be marked on the y- axis (decimal 0.3…… allowed) There must be a curve drawn (even incorrectly) to achieve this mark. **Do not** accept intersections unless they are marked on the graph in the correct place.
(b)	M1	For the use of the power rule with logs to obtain $2\log_a 4$ from $\log_a 16$ OR $4\log_a 2$ from $\log_a 16$
	dM1	A **correct** rearrangement of their equation to obtain for example $a^{\frac{1}{6}} = 4$ or $a^{\frac{1}{12}} = 2$ or even of the type $a^1 = 64^2$ There are many ways of completing this. Look for a correct value for the corresponding correct power of a. **This mark is dependent on the first M mark**. Look for correct log work for their values.
	A1	For $a = 4096$
ALT	M1	For use of the correct log rule to obtain $\log_a 64$
	dM1	A correct rearrangement of their equation to obtain $a^{\frac{1}{2}} = 64$ **This mark is dependent on the first M mark**. Look for correct log work for their values.
	A1	For $a = 4096$
(c)	M1	For a fully correct change of base of log to either base 2 or base q
	M1	For an attempt to rearrange and form a 3TQ quadratic. Allow one arithmetical slip only.
	A1	For the correct 3TQ.
	dM1	**Dependent on the second M mark** – an acceptable attempt to solve the quadratic equation – see general guidance.
	M1	For a correct application using either of their solutions moving from log form to exponential form. **Note this is an independent mark.**
	A1	**Both** correct solutions.

Type MS · Log equations: combine into one log, then remove the log

Answer 4 of 16 · Question 7 part b · 3 marks

2022 JUN · Paper 01 (regional R) · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 5 of 16 · Question 7 part c · 3 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

7(a)	4	B1 [1]
(b)	**Working in $\log_2$** $2\log_4 x = \frac{2\log_2 x}{\log_2 4} = \frac{2\log_2 x}{2} = [\log_2 x]$ $\log_2 16 + \frac{2\log_2 x}{2} = \log_2 y$ $\log_2 16x = \log_2 y$ OR $\log_2\left(\frac{x}{y}\right) = -4 \Rightarrow \frac{x}{y} = 2^{-4}$ $y = 16x^*$	M1 M1 M1 A1 cso [4]
	ALT	
	Working in $\log_4$ $\log_2 y = \frac{\log_4 y}{\log_4 2} = \frac{\log_4 y}{\frac{1}{2}} = 2\log_4 y = [\log_4 y^2]$ $\log_4 256 + \log_4 x^2 = \log_4 y^2$ $\log_4\left(256x^2\right) = \log_4 y^2$ OR $2\log_4\left(\frac{y}{x}\right) = 4 \Rightarrow \frac{y}{x} = 4^2$ $256x^2 = y^2 \Rightarrow y = 16x^*$	[M1 M1 M1 A1]
(c)	$16x = 4x + 5$ $16x = 4x + 5 \Rightarrow 12x = 5 \Rightarrow x = ...$ $x = \frac{5}{12}$	B1 M1 A1 [3]
		Total 8 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 5 of 16 · Question 7 part c · 3 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Alt – working in $\log_4$		
	M1	For an attempt to change the base of $\log_2 y$ to base 4 using $\log_a y = \frac{\log_b y}{\log_b a}$ $\log_2 y = \frac{\log_4 y}{\log_4 2}\left[= \frac{\log_4 y}{\frac{1}{2}} = 2\log_4 y\right]$
	M1	For dealing with the indices **and** writing $4 = \log_4 256$ $\log_4 256 + \log_4 x^2 = \log_4 y^2$
	M1	Uses $\log A + \log B = \log AB$ to correctly combine the logs $\log_4\left(256x^2\right) = \log_4 y^2$ OR Uses $\log A - \log B = \log\frac{A}{B}$ to correctly combine the logs and removes logs $2\log_4\left(\frac{y}{x}\right) = 4$ **and** $\frac{y}{x} = 4^2$ (this approach will score the second and third M marks at this stage)
	A1	For correctly obtaining $y = 16x$*
(c)	**B1**	For writing down $16x = 4x + 5$
	M1	For an attempt to solve the equation $16x = 4x + 5 \Rightarrow 12x = 5 \Rightarrow x = ...$
	A1	For $x = \frac{5}{12}$
	Note: This is a 'hence' question. Condone candidates working without using given results, but the first mark is not awarded until the candidate reaches $16x = 4x + 5$	

Type MS · Log equations: combine into one log, then remove the log

Answer 5 of 16 · Question 7 part c · 3 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 6 of 16 · Question 9 part i · 3 marks

2025 JUN · Paper 02 (regional R) · extract 1 of 3 · find part i in this mark scheme extract

Number		
9 (i)	$\left(3(\log_a 9+\log_a 27)=1\right)$	
	$3(2\log_a 3+3\log_a 3)=1$ oe eg $2\log_a 3+3\log_a 3=\frac{1}{3}$	M1
	$\log_a 3=\frac{1}{15}$	
	$a^{\frac{1}{15}}=3$ or $\log_3 a=15$	M1
	$(a=)3^{15}$	A1
	ALT	[3]
	$3(\log_a(9\times 27)=1$ oe eg $3(\log_a(3^5))=1$	M1
	$\log_a(9\times 27)=\frac{1}{3}$ oe eg $\log_a(3^5)=\frac{1}{3}$	
	$a^{\frac{1}{3}}=9\times 27$	M1
	$(a=)3^{15}$	A1
		[3]
(ii)	$\left(\log_4 p+\log_p 256=-4\right)$	
	$\log_4 p+\frac{\log_4 256}{\log_4 p}=-4$ or $\frac{\log_p p}{\log_p 4}+\log_p 256=-4$	M1
	$(\log_4 p)^2+\log_4 256=-4\log_4 p$ or $(\log_p p)+(\log_p 256)(\log_p 4)=-4\log_p 4$	dM1
	$(\log_4 p)^2+4\log_4 p+4(=0)$ or $4(\log_p 4)^2+4\log_p 4+1(=0)$	A1
	$((\log_4 p)+2)^2(=0)$ or $((2\log_p 4)+1)^2(=0)$	M1
	$\log_4 p=-2\rightarrow p=$ or $\log_p 4=-\frac{1}{2}\rightarrow p=$	M1
	$(p=)\frac{1}{16}$ oe	A1
		[6]
		Total 9 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 6 of 16 · Question 9 part i · 3 marks

2025 JUN · Paper 02 (regional R) · extract 2 of 3 · find part i in this mark scheme extract

Part	Mark	Additional Guidance
When marking (i) stick to 1 ALTERNATIVE only – eg it's not possible to gain M2 for an application of a power rule and an addition rule with logs. **Look out for equivalent working, particularly the use of** $\log_{a^b} c = \frac{1}{b}\log_a c$ **Send any work that you may think uses a different method worth partial marks to Review**		
(i)	M1	For correct use of the rule $\log_a A^b = b\log_a A$ This mark may be awarded anywhere the power rule is correctly used **once** in working, in either direction, even if there's other contradictory work.
	M1	For a rearrangement of $3(b\log_a 3 + c\log_a 3) = 1$ to $a^{\frac{1}{3R}} = 3$ or $\log_3 a = 3R \quad R > 1$ seen anywhere in the working, even if there's other contradictory work. Allow errors in rearrangement. May be implied by any correct value for a not in the required form eg 14348907, $\left(3^5\right)^3$, $\left(3^3\right)^5$, $9^{7.5}$
	A1	$(a =) 3^{15}$
ALT	M1	For correct use the rule $\log_a A + \log_a B = \log_a AB$ This mark may be awarded anywhere the addition rule is correctly used **once** in working, in either direction, even if there's other contradictory work.
	M1	For a rearrangement of $3(\log_a (C \times D)) = 1$ to $a^{\frac{1}{3}} = CD$ seen anywhere in the working, even if there's other contradictory work. Allow errors in rearrangement. May be implied by any correct value for x not in the required form eg 14348907, $\left(3^5\right)^3$, $\left(3^3\right)^5$, $9^{7.5}$
	A1	$(a =) 3^{15}$
(ii)	M1	Correct use of change of base of logs. This mark may be awarded anywhere a change of base of logs is used correctly **once** in working, even if there's other contradictory work.
	dM1	Rearranges their equation to a 3 term quadratic (= 0). Allow errors in rearrangement so long as reach an expression of the form $p(\log_4 p)^2 + q\log_4 p + r = 0$ or $s(\log_p 4)^2 + t\log_p 4 + u = 0 \quad p,q,r,s,t,u \neq 0$
	A1	Correct 3 term quadratic(= 0) as shown.
	M1	A minimally acceptable attempt to solve their 3 term quadratic, see general guidance. Students may have used an alternative variable here for $\log_4 p$ or $\log_p 4$ The mark is for a minimally acceptable attempt to solve a 3TQ – see general guidance. The correct value from the correct quadratic implies this mark, otherwise method must be shown.
	M1	Correctly coverts their log equation into a value for p. ie $\log_4 p = F \rightarrow p =$ or $\log_p 4 = G \rightarrow p =$
	A1	For $(p =) \frac{1}{16}$ oe

Type MS · Log equations: combine into one log, then remove the log

Answer 6 of 16 · Question 9 part i · 3 marks

2025 JUN · Paper 02 (regional R) · extract 3 of 3 · find part i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 7 of 16 · Question 3 part b · 6 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

3 (a)	$\log_3 9 = 2$	B1 [1]
(b)	$\log_3 9t = \log_9\left(\frac{12}{t}\right)^2 + 2 \Rightarrow \log_3 9 + \log_3 t = 2\left(\log_9 12 - \log_9 t\right) + 2$	M1M1
	$\log_3 9 + \log_3 t = 2\left(\frac{\log_3 12}{\log_3 9} - \frac{\log_3 t}{\log_3 9}\right) + 2$	M1
	$\Rightarrow \log_3 9 + \log_3 t = \log_3 12 - \log_3 t + 2$	
	$\Rightarrow 2\log_3 t = \log_3 12 \Rightarrow \log_3 t^2 = \log_3 12$	A1
	$\Rightarrow t^2 = 12 \Rightarrow t = 2\sqrt{3}$	M1A1 [6]
		Total 7 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 7 of 16 · Question 3 part b · 6 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 8 of 16 · Question 7 part b · 4 marks

2022 JAN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

7(a)	4	B1 [1]
(b)	**Working in $\log_2$** $2\log_4 x = \frac{2\log_2 x}{\log_2 4} = \frac{2\log_2 x}{2} = [\log_2 x]$ $\log_2 16 + \frac{2\log_2 x}{2} = \log_2 y$ $\log_2 16x = \log_2 y$ OR $\log_2\left(\frac{x}{y}\right) = -4 \Rightarrow \frac{x}{y} = 2^{-4}$ $y = 16x^*$	M1 M1 M1 A1 cso [4]
	ALT	
	Working in $\log_4$ $\log_2 y = \frac{\log_4 y}{\log_4 2} = \frac{\log_4 y}{\frac{1}{2}} = 2\log_4 y = [\log_4 y^2]$ $\log_4 256 + \log_4 x^2 = \log_4 y^2$ $\log_4\left(256x^2\right) = \log_4 y^2$ OR $2\log_4\left(\frac{y}{x}\right) = 4 \Rightarrow \frac{y}{x} = 4^2$ $256x^2 = y^2 \Rightarrow y = 16x^*$	[M1 M1 M1 A1]
(c)	$16x = 4x + 5$ $16x = 4x + 5 \Rightarrow 12x = 5 \Rightarrow x = ...$ $x = \frac{5}{12}$	B1 M1 A1 [3]
		Total 8 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 8 of 16 · Question 7 part b · 4 marks

2022 JAN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Alt – working in $\log_4$		
	M1	For an attempt to change the base of $\log_2 y$ to base 4 using $\log_a y = \frac{\log_b y}{\log_b a}$ $\log_2 y = \frac{\log_4 y}{\log_4 2}\left[= \frac{\log_4 y}{\frac{1}{2}} = 2\log_4 y\right]$
	M1	For dealing with the indices **and** writing $4 = \log_4 256$ $\log_4 256 + \log_4 x^2 = \log_4 y^2$
	M1	Uses $\log A + \log B = \log AB$ to correctly combine the logs $\log_4\left(256x^2\right) = \log_4 y^2$ OR Uses $\log A - \log B = \log\frac{A}{B}$ to correctly combine the logs and removes logs $2\log_4\left(\frac{y}{x}\right) = 4$ **and** $\frac{y}{x} = 4^2$ (this approach will score the second and third M marks at this stage)
	A1	For correctly obtaining $y = 16x$*
(c)	**B1**	For writing down $16x = 4x + 5$
	M1	For an attempt to solve the equation $16x = 4x + 5 \Rightarrow 12x = 5 \Rightarrow x = ...$
	A1	For $x = \frac{5}{12}$
	Note: This is a 'hence' question. Condone candidates working without using given results, but the first mark is not awarded until the candidate reaches $16x = 4x + 5$	

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Type MS · Log equations: combine into one log, then remove the log

Answer 8 of 16 · Question 7 part b · 4 marks

2022 JAN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 9 of 16 · Question 9 part b · 4 marks

2024 JUN · Paper 01 · extract 1 of 3 · find part b in this mark scheme extract

9 (a)	$\left(\log_a 8=\frac{3}{4}\Rightarrow\right)a^{\frac{3}{4}}=8\left(\Rightarrow a=\left(\sqrt[3]{8}\right)^4\right)=16$	M1A1 [2]
(b)	$\left(3x\log_2 x-4\log_{16}8+6x\log_4 8-\log_2 x=\right)3x\log_2 x-\frac{4\log_2 8}{\log_2 16}+\frac{6x\log_2 8}{\log_2 4}-\log_2 x$ $\Rightarrow 3x\log_2 x-\log_2 8+3x\log_2 8-\log_2 x$ $=(3x-1)\log_2 x+(3x-1)\log_2 8$ or $3x\log_2 8x-(1)\log_2 8x$ $\Rightarrow(3x-1)\log_2 8x$ or $\log_2(8x)^{3x}+\log_2(8x)^{-1}=\log_2(8x)^{3x-1}$ *	M1 M1 M1A1 cso [4]
ALT	$\left(3x\log_2 x-4\log_{16}8+6x\log_4 8-\log_2 x=\right)3x\log_2 x-\frac{4\log_2 8}{\log_2 16}+\frac{6x\log_2 x}{\log_2 4}-\log_2 x$ $\Rightarrow 3x\log_2 x-\log_2 8+3x\log_2 8-\log_2 x=\log_2 x^{3x}-\log_2 8+\log_2 8^{3x}-\log_2 x$ $\left(\Rightarrow\log_2(8x)^{3x}-\log_2 8x\Rightarrow\right)\log_2\left(\frac{(8x)^{3x}}{8x}\right)$ or $\log_2\left((8x)^{3x}\times(8x)^{-1}\right)$ or $\log_2 x^{3x-1}8^{3x-1}=\log_2(8x)^{3x-1}$ *	M1 M1 M1A1 cso [4]
"Box 3" of part b We will see unanticipated methods once live marking begins. If the answer is correct and **there is no incorrect working**, check the work carefully, to ascertain if they've shown enough steps to demonstrate use of the three main log laws this question tests and award full marks. If in any doubt at all – the response **MUST** be sent to review. Other than this exception, please mark to the following rules. **Also use these rules if students don't gain the 2nd or 3rd M under the main or ALT schemes.** • M1 for any correct change of base to base 2 • M1 for any two correct applications of the power law or for $ax\log_2 8x+b\log_2 8x\Rightarrow\log_2(8x)^{ax+b}$ or $(ax+b)\log_2 8x\Rightarrow\log_2(8x)^{ax+b}$ $a,b\neq 0$ • M1 for any two correct applications of the addition or subtraction law In each case – ignore any incorrect working. **Poor or incorrect bracketing may not be recovered in this question**. (general principle of marking is usually that it can).	
(c)	$\left[\log_2(8x)^{3x-1}=0\Rightarrow\log_2(8x)^{3x-1}=\log_2 8^0 \text{ or } (3x-1)\log_2(8x)=0\right]$ $\Rightarrow 3x-1=0\Rightarrow x=\frac{1}{3}$ $\Rightarrow 8x=1\Rightarrow x=\frac{1}{8}$	M1A1 A1 [3]
		Total 9 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 9 of 16 · Question 9 part b · 4 marks

2024 JUN · Paper 01 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For undoing the log to obtain $a^{\frac{3}{4}}=8$ or $(a=)8^{\frac{4}{3}}$
	A1	For $a = 16$ M1 A1 may be awarded for just seeing $a = 16$ unless from a string of incorrect work. Poor notation may be seen.
(b)	M1	For correctly changing the base to base 2 on at least one log, anywhere in their work. This may be implied and the mark awarded for eg $4\log_{16}8 \Rightarrow \frac{4\log_2 8}{\log_2 16}$ or $\frac{4\log_2 8}{4}$ or $\log_2 8$ or $\frac{12}{4}$ or 3 or $6x\log_4 8 \Rightarrow \frac{6x\log_2 8}{\log_2 4}$ or $\frac{6x\log_2 8}{2}$ or $3x\log_2 8$ or $9x$
	M1	For $(3x-1)\log_2 x+(3x-1)\log_2 8$
	M1	For correctly applying the power log law to their expression of the form $(ax+b)\log_2 8x \quad a,b\neq 0$ In general, poor bracketing may be recovered, but as this question is a show question, it generally cannot be recovered eg do not accept $8x^{3x}$
	A1 cso	For the correct expression, **minimum steps as shown** with no errors or omissions.
ALT	M1	For correctly changing the base to base 2 on at least one log. This may be implied and the mark awarded for $4\log_{16}8 \Rightarrow \frac{4\log_2 8}{\log_2 16}$ or $\frac{4\log_2 8}{4}$ or $\log_2 8$ or $\frac{12}{4}$ or 3 or $6x\log_4 8 \Rightarrow \frac{6x\log_2 8}{\log_2 4}$ or $\frac{6x\log_2 8}{2}$ or $3x\log_2 8$ or $9x$
	M1	For $\log_2 x^{(3x)}-\log_2 8+\log_2 8^{(3x)}-\log_2 x$
	M1	For correctly applying the subtraction law to an expression of the form (see the MS for minimum steps) $\log_2 x^{(ax)}-\log_2 8^b+\log_2 8^{(ax)}-\log_2 x^b$ The step in brackets doesn't need to be shown. Do not permit only eg $\log_2\left(x^{3x}\div 8\times 8^{3x}\div x\right)$ as a sufficient minimum step. In general, poor bracketing may be recovered, but as this question is a show question, it generally cannot be recovered eg do not accept $8x^{3x}\times 8x^{-1}$
	A1	For the correct expression, **minimum steps as shown** with no errors or omissions. .
	SC4	For working on both sides **with no errors** and achieving lhs = rhs Note, this is an exception to what we normally allow in a show that question (ie working on both sides until agreement)
(c)	M1	For setting $3x-1=0$ or $8x=1$ This mark may be implied by a correct answer.
	A1	For either $x=\frac{1}{3}$ or $\frac{1}{8}$
	A1	For both $x=\frac{1}{3}$ and $\frac{1}{8}$

Type MS · Log equations: combine into one log, then remove the log

Answer 9 of 16 · Question 9 part b · 4 marks

2024 JUN · Paper 01 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 10 of 16 · Question 9 part c · 3 marks

2024 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

9 (a)	$\left(\log_a 8=\frac{3}{4}\Rightarrow\right)a^{\frac{3}{4}}=8\left(\Rightarrow a=\left(\sqrt[3]{8}\right)^4\right)=16$	M1A1 [2]
(b)	$\left(3x\log_2 x-4\log_{16}8+6x\log_4 8-\log_2 x=\right)3x\log_2 x-\frac{4\log_2 8}{\log_2 16}+\frac{6x\log_2 8}{\log_2 4}-\log_2 x$ $\Rightarrow 3x\log_2 x-\log_2 8+3x\log_2 8-\log_2 x$ $=(3x-1)\log_2 x+(3x-1)\log_2 8$ or $3x\log_2 8x-(1)\log_2 8x$ $\Rightarrow(3x-1)\log_2 8x$ or $\log_2(8x)^{3x}+\log_2(8x)^{-1}=\log_2(8x)^{3x-1}$ *	M1 M1 M1A1 cso [4]
ALT	$\left(3x\log_2 x-4\log_{16}8+6x\log_4 8-\log_2 x=\right)3x\log_2 x-\frac{4\log_2 8}{\log_2 16}+\frac{6x\log_2 x}{\log_2 4}-\log_2 x$ $\Rightarrow 3x\log_2 x-\log_2 8+3x\log_2 8-\log_2 x=\log_2 x^{3x}-\log_2 8+\log_2 8^{3x}-\log_2 x$ $\left(\Rightarrow\log_2(8x)^{3x}-\log_2 8x\Rightarrow\right)\log_2\left(\frac{(8x)^{3x}}{8x}\right)$ or $\log_2\left((8x)^{3x}\times(8x)^{-1}\right)$ or $\log_2 x^{3x-1}8^{3x-1}=\log_2(8x)^{3x-1}$ *	M1 M1 M1A1 cso [4]
"Box 3" of part b We will see unanticipated methods once live marking begins. If the answer is correct and **there is no incorrect working**, check the work carefully, to ascertain if they've shown enough steps to demonstrate use of the three main log laws this question tests and award full marks. If in any doubt at all – the response **MUST** be sent to review. Other than this exception, please mark to the following rules. **Also use these rules if students don't gain the 2nd or 3rd M under the main or ALT schemes.** • M1 for any correct change of base to base 2 • M1 for any two correct applications of the power law or for $ax\log_2 8x+b\log_2 8x\Rightarrow\log_2(8x)^{ax+b}$ or $(ax+b)\log_2 8x\Rightarrow\log_2(8x)^{ax+b}$ $a,b\neq 0$ • M1 for any two correct applications of the addition or subtraction law In each case – ignore any incorrect working. **Poor or incorrect bracketing may not be recovered in this question**. (general principle of marking is usually that it can).	
(c)	$\left[\log_2(8x)^{3x-1}=0\Rightarrow\log_2(8x)^{3x-1}=\log_2 8^0\right.$ or $\left.(3x-1)\log_2(8x)=0\right]$ $\Rightarrow 3x-1=0\Rightarrow x=\frac{1}{3}$ $\Rightarrow 8x=1\Rightarrow x=\frac{1}{8}$	M1A1 A1 [3]
		Total 9 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 10 of 16 · Question 9 part c · 3 marks

2024 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For undoing the log to obtain $a^{\frac{3}{4}}=8$ or $(a=)8^{\frac{4}{3}}$
	A1	For $a = 16$ M1 A1 may be awarded for just seeing $a = 16$ unless from a string of incorrect work. Poor notation may be seen.
(b)	M1	For correctly changing the base to base 2 on at least one log, anywhere in their work. This may be implied and the mark awarded for eg $4\log_{16}8 \Rightarrow \frac{4\log_2 8}{\log_2 16}$ or $\frac{4\log_2 8}{4}$ or $\log_2 8$ or $\frac{12}{4}$ or 3 or $6x\log_4 8 \Rightarrow \frac{6x\log_2 8}{\log_2 4}$ or $\frac{6x\log_2 8}{2}$ or $3x\log_2 8$ or $9x$
	M1	For $(3x-1)\log_2 x+(3x-1)\log_2 8$
	M1	For correctly applying the power log law to their expression of the form $(ax+b)\log_2 8x \quad a,b\neq 0$ In general, poor bracketing may be recovered, but as this question is a show question, it generally cannot be recovered eg do not accept $8x^{3x}$
	A1 cso	For the correct expression, **minimum steps as shown** with no errors or omissions.
ALT	M1	For correctly changing the base to base 2 on at least one log. This may be implied and the mark awarded for $4\log_{16}8 \Rightarrow \frac{4\log_2 8}{\log_2 16}$ or $\frac{4\log_2 8}{4}$ or $\log_2 8$ or $\frac{12}{4}$ or 3 or $6x\log_4 8 \Rightarrow \frac{6x\log_2 8}{\log_2 4}$ or $\frac{6x\log_2 8}{2}$ or $3x\log_2 8$ or $9x$
	M1	For $\log_2 x^{(3x)}-\log_2 8+\log_2 8^{(3x)}-\log_2 x$
	M1	For correctly applying the subtraction law to an expression of the form (see the MS for minimum steps) $\log_2 x^{(ax)}-\log_2 8^b+\log_2 8^{(ax)}-\log_2 x^b$ The step in brackets doesn't need to be shown. Do not permit only eg $\log_2\left(x^{3x}\div 8\times 8^{3x}\div x\right)$ as a sufficient minimum step. In general, poor bracketing may be recovered, but as this question is a show question, it generally cannot be recovered eg do not accept $8x^{3x}\times 8x^{-1}$
	A1	For the correct expression, **minimum steps as shown** with no errors or omissions. .
	SC4	For working on both sides **with no errors** and achieving lhs = rhs Note, this is an exception to what we normally allow in a show that question (ie working on both sides until agreement)
(c)	M1	For setting $3x-1=0$ or $8x=1$ This mark may be implied by a correct answer.
	A1	For either $x=\frac{1}{3}$ or $\frac{1}{8}$
	A1	For both $x=\frac{1}{3}$ and $\frac{1}{8}$

Type MS · Log equations: combine into one log, then remove the log

Answer 10 of 16 · Question 9 part c · 3 marks

2024 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 11 of 16 · Question 7 part whole · 5 marks

2020 JAN · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

7	$\frac{\log_7 x^2}{\log_7 49}$ $\log_7\left(\frac{8x^2-6x+3}{x}\right)$, $\log_7 2^3$ $\frac{8x^2-6x+3}{x} = 2^3$ $8x^2 - 14x + 3 = 0$ $(4x-1)(2x-3) = 0$ $x = \frac{1}{4}, \frac{3}{2}$	B1 M1 A1 M1 A1 **[5]**

Mark	Additional Guidance	
B1	For changing the base of the log either to base 7 or base 49	
	$\log_{49} x^2 = \frac{\log_7 x^2}{\log_7 49} = \frac{\log_7 x^2}{2}$ OR $\log_{49} x^2 = \frac{2\log_7 x}{\log_7 49} = \log_7 x$	$\log_7\left(8x^2-6x+3\right) = \frac{\log_{49}\left(8x^2-6x+3\right)}{\log_{49} 7}$ $= 2\log_{49}\left(8x^2-6x+3\right)$ **AND** $\log_7 2 = \frac{\log_{49} 2}{\log_{49} 7} = 2\log_{49} 2$
M1	For combining the LHS together into one log and dealing with the powers on both sides	
	$\left[\frac{1}{2}\log_7 x^2 = \log_7 x\right] \Rightarrow$ $\log_7\left(\frac{8x^2-6x+3}{x}\right)$, $\log_7 2^3$	$\log_{49}\left(\frac{\left[8x^2-6x+3\right]^2}{x^2}\right)$, $\log_{49} 2^6$
dM1	For forming a 3TQ with their expressions which must have come from an acceptable attempt to deal with the logs **This is an A mark in Epen**	
	$8x^2-14x+3=0$	$\left(8x^2-6x+3\right)^2 = 64x^2 \Rightarrow$ $8x^2-6x+3 = \pm 8x \Rightarrow 8x^2-14x+3=0$ If this method is used they must reject the negative root of $64x^2$ (i.e $-8x$) because it will form a quadratic equation with no real roots. $\left\{8x^2+2x+3=0 \Rightarrow b^2-4ac=-92\right\}$
dM1	For attempting to solve **their** 3TQ $8x^2-14x+3=(4x-1)(2x-3)=0 \Rightarrow x=...,...$	
A1	$x=\frac{3}{2}$, $\frac{1}{4}$	

Type MS · Log equations: combine into one log, then remove the log

Answer 11 of 16 · Question 7 part whole · 5 marks

2020 JAN · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 12 of 16 · Question 10 part ii · 9 marks

2025 JUN · Paper 01 · extract 1 of 3 · find part ii in this mark scheme extract

10(i)	$\log_4(6y-5)=3 \Rightarrow 6y-5=64 \Rightarrow y=\frac{23}{2}$ oe	M1A1 [2]
(ii)	$\log_4(4-3x)^2-\frac{\log_4(x^2-5)}{\log_4 2}=3$ or $\frac{\log_2(4-3x)^2}{\log_2 4}-\log_2(x^2-5)=3$	M1
	$\frac{1}{2}\log_4(4-3x)^2-\log_4(x^2-5)=\frac{3}{2}$ or $\frac{1}{2}\log_2(4-3x)^2-\log_2(x^2-5)=3$	B1
	$\log_4(4-3x)-\log_4(x^2-5)=\frac{3}{2}$ or $\log_2(4-3x)-\log_2(x^2-5)=3$	M1
	$\log_4\left(\frac{4-3x}{x^2-5}\right)=\frac{3}{2}$ or $\log_2\left(\frac{4-3x}{x^2-5}\right)=3$	M1
	$\frac{4-3x}{x^2-5}=4^{\frac{3}{2}}=8$ or $\frac{4-3x}{x^2-5}=2^3=8$	M1
	$8x^2+3x-44=0$	A1
	$x=\frac{-3\pm\sqrt{3^2-4\times8\times-44}}{2\times8}=\frac{-3\pm\sqrt{1417}}{16} \Rightarrow \{x=2.1651...$ or $-2.540...\}$	M1A1
	$\therefore x=-2.54$ (awrt)	A1 [9]
		Total 11 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 12 of 16 · Question 10 part ii · 9 marks

2025 JUN · Paper 01 · extract 2 of 3 · find part ii in this mark scheme extract

Part	Mark	Additional Guidance
(i)	M1	For a complete correct method to find the value of y
	A1	For the correct value
For part (ii) award the marks when you see them		
(ii)	M1	For change of base correctly to either 4 or 2
	B1	For replacing $\log_2 4$ with 2 or replacing $\log_4 2$ with $\frac{1}{2}$
	M1	For applying the power law correctly e.g.: $\frac{1}{2}\log_4(4-3x)^2 \Rightarrow \log_4(4-3x)$ or $\frac{1}{2}\log_2(4-3x)^2 \Rightarrow \log_2(4-3x)$
	M1	For applying the subtraction/addition law correctly
	M1	For correctly removed logs
	A1	For the correct 3TQ, "=0 "could be implied by later work
	M1	For a correct clear method to solve their 3TQ, allow one/two solutions from a calculator correct to 1dp (awrt) from a **correct** 3TQ
	A1	For two correct solutions, **exact** (allow embedded in a quadratic formula) or $x = 2.17$(awrt), $x = -2.54$(awrt) or JUST $x = -2.54$(awrt) seen only, from a correct 3TQ, scores M1A1A1(rejected 2.17 is implied)
	A1	For the correct solution. Accept awrt -2.54 The positive root must be rejected as it is not feasible.

Type MS · Log equations: combine into one log, then remove the log

Answer 12 of 16 · Question 10 part ii · 9 marks

2025 JUN · Paper 01 · extract 3 of 3 · find part ii in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 13 of 16 · Question 5 part whole · 7 marks

2021 NOV · Paper 01 · extract 1 of 2 · find part whole in this mark scheme extract

5	$\log_3\sqrt{x-5}+\log_9(x+3)-1=0$ $\frac{1}{2}\log_3(x-5)+\frac{\log_3(x+3)}{\log_3 9}=1\Rightarrow\left(\frac{1}{2}\log_3(x-5)+\frac{\log_3(x+3)}{2}=1\right)$ $\log_3(x-5)+\log_3(x+3)=2\Rightarrow\log_3\left[(x-5)(x+3)\right]=2$ $\Rightarrow(x-5)(x+3)=3^2\Rightarrow x^2-2x-24=0$ $(x+4)(x-6)=0\Rightarrow x=6$ (reject $x=-4$)	M1M1 M1 M1A1 dM1A1 [7]
		Total 7 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 13 of 16 · Question 5 part whole · 7 marks

2021 NOV · Paper 01 · extract 2 of 2 · find part whole in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 14 of 16 · Question 5 part a · 2 marks

2020 JAN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

M1	$f'(2) = 5$ leading to an equation in a and b
A1	$4a + 2b + 34 = 0$ and $4a + b + 19 = 0$
M1	Solving simultaneously
A1	$b = -15,\ a = -1$
(b)	
M1	Dividing by $x - 2$ to obtain a 3TQ
M1	Factorising the 3TQ
A1	All 3 terms correct
(c)	
B2 ft	$x = 2,\ -3,\ \frac{3}{2}$ (**B1** for 2 correct)

Type MS · Log equations: combine into one log, then remove the log

Answer 14 of 16 · Question 5 part a · 2 marks

2020 JAN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks
5 (a)	$\log_4 32 = \frac{\log_2 32}{\log_2 4} = \frac{5}{2}*$ or $\log_4 32 = \log_4 4^{\frac{5}{2}} = \frac{5}{2}*$ or $\log_4 32 = \log_{2^2} 2^5 = \frac{5}{2}*$ **ALT** $\log_4 32 = a \quad \Rightarrow 4^a = 32 \quad \Rightarrow a = \frac{5}{2} \quad *$	M1A1cso [2] {M1}{A1} cso [2]
(b)	$\log_2 x - \log_4 32 + \frac{1}{4}\log_x 16 = 0$ Let $\log_2 x = y$ $y - \frac{5}{2} + \frac{1}{4}\left(\frac{\log_2 16}{\log_2 x}\right) = 0$ or $y - \frac{5}{2} + \frac{1}{\log_2 x} = 0$ $\Rightarrow y - \frac{5}{2} + \frac{1}{y} = 0$ $\Rightarrow 2y^2 - 5y + 2 = 0$ $\Rightarrow (2y-1)(y-2) = 0$ $\Rightarrow y = \log_2 x = \frac{1}{2}$ or 2 $\Rightarrow x = 2^{\frac{1}{2}} = \sqrt{2}$ and $x = 2^2 = 4$	M1 M1A1 M1 M1 M1A1 [7]
		Total 9 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 15 of 16 · Question 6 part i · 4 marks

2024 NOV · Paper 02 · extract 1 of 3 · find part i in this mark scheme extract

Number		
6 (i)	$5(2\log_b 3+\log_b 3)=3$	M1
	$\left(3\log_b 3=\frac{3}{5}\right)\Rightarrow \log_b 3=\frac{1}{5}$	M1
	$b^{\frac{1}{5}}=3$	M1
	$b=243$ or 3^5	A1
	ALT	
	$5\log_b 27=3$	M1
	$\log_b 27=\frac{3}{5}$	M1
	$b^{\frac{3}{5}}=27$	M1
	$b=243$ or $b=27^{\frac{5}{3}}$	A1 [4]
(ii)	$3\log_3 x+3\frac{\log_3 27}{\log_3 x}=,28$	M1,B1
	$3(\log_3 x)^2+9=28\log_3 x$	dM1
	$3(\log_3 x)^2-28\log_3 x+9(=0)\Rightarrow(3\log_3 x-1)(\log_3 x-9)(=0)$	ddM1
	$(\log_3 x=)\frac{1}{3}$ and $(\log_3 x=)9$	A1
	$x=3^{\frac{1}{3}},3^9$ oe exact form	M1 A1 [7]
		Total 11 marks

Type MS · Log equations: combine into one log, then remove the log

Answer 15 of 16 · Question 6 part i · 4 marks

2024 NOV · Paper 02 · extract 2 of 3 · find part i in this mark scheme extract

Part	Mark	Additional Guidance
(i)	M1	Rewrites 9 as 3^2 and correctly uses the power log rule as shown.
	M1	Collects terms as rearranges to give $\log_b p = r$, allow 1 error in rearrangement.
	M1	Correctly converts their log equation to an exponential.
	A1	Cao
ALT	M1	Uses the log addition law correctly to combine into a single log.
	M1	Collects terms as rearranges to give $\log_b p = r$, allow 1 error in rearrangement.
	M1	Rearranges to the form $b^{\frac{p}{q}} = r$, allow 1 error in rearrangement.
	A1	Cao
(ii)	B1	Replacement of $8\log_4 128 = 28$, wherever we see
	M1	For a correct change of the same base of logs on the lhs of the equation e.g. $3\log_3 x + 3\dfrac{\log_3 27}{\log_3 x} = 28$ or $\dfrac{3}{\log_x 3} + 3\log_x 27 = 28$ Allow slips on the coefficients
	dM1	For forming a 3-term quadratic in any form e.g. $3(\log_3 x)^2 + 9 = 28\log_3 x$ or $3 + 9(\log_x 3)^2 = 28(\log_x 3)$
	ddM1	For solving their 3-term quadratic using a valid method – see general guidance. (Correct answers from a correct 3TQ scores this mark)
	A1	For $(\log_3 x =)\dfrac{1}{3}$ and $(\log_3 x =)9$ or $(\log_x 3) = 3$ and $(\log_x 3) = \dfrac{1}{9}$
	M1	For correctly rearranging either of their equations, removes logs to find $x =$
	A1	For both correct values oe exact form. E.g. $x = 19683, x = \sqrt[3]{3}$ **isw**

Type MS · Log equations: combine into one log, then remove the log

Answer 15 of 16 · Question 6 part i · 4 marks

2024 NOV · Paper 02 · extract 3 of 3 · find part i in this mark scheme extract

Question number	Scheme	Marks

Type MS · Log equations: combine into one log, then remove the log

Answer 16 of 16 · Question 4 part whole · 7 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part whole in this mark scheme extract

4(i)	$\dfrac{16}{\log_4 r} = \log_4 r \Rightarrow 16 = (\log_4 r)^2 \Rightarrow \log_4 r = \pm 4$	M1	
	$r = 4^4 = 256 \quad \text{or } r = 4^{-4} = \dfrac{1}{256}$	A1	(2)
(ii)	$\log_5 9 + \log_5 12 + \log_5 15 + \log_5 18 = \log_5 (9\times12\times15\times18) = \log_5 29\,160$	M1	
	$1 + \log_5 x + \log_5 x^2 = \log_5 5 + \log_5 x + \log_5 x^2 = \log_5 5x^3$	M1A1	
	$5x^3 = 29\,160$	dM1	
	$x = 18$	A1	(5) [7]
ALT 1			
	$\text{LHS} = \log_5 29\,160$	M1	
	$\text{RHS} = 1 + \log_5 x^3$	M1	
	$\left(\dfrac{\log_{10} 29\,160}{\log_{10} 5}\right) = 6.387\ldots \left(= \log_5 x^3 + 1\right)$	A1	
	$5.387\ldots = 3\log_5 x$	dM1	
	$\log_5 x = 1.795\ldots$		
	$x = 18$	A1	
ALT 2	$\text{LHS} = \log_5 29\,160$	M1	
	$\text{RHS} = \log_5 5 + \log_5 x^3$	M1A1	
	$\log_5 29\,160 = \log_5 5 + \log_5 5832$		
	$5832 = x^3$	dM1	
	$x = 18$	A1	
ALT 3	$\text{LHS} = \log_5 5832 + \log_5 5$	M1	
	$\text{RHS} = 1 + \log_5 x^3$	M1	
	$\text{LHS} = \log_5 5832 + 1$	A1	
	$\log_5 5832 = \log_5 x^3$		
	$5832 = x^3$	dM1	
	$x = 18$	A1	
ALT 4	$\log_5 29\,160 - \log_5 x^3 = 1$	M1M1	
	$\log_5 \dfrac{29\,160}{x^3} = 1$	A1	
	$\dfrac{29\,160}{x^3} = 5 \Rightarrow x^3 = 5832$	dM1	
	$x = 18$	A1	

Type MS · Log equations: combine into one log, then remove the log

Answer 16 of 16 · Question 4 part whole · 7 marks

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(i)	
M1	Change base (can have base 4 or base r provided the same for both logs), multiply to remove the fraction and solve to $\log_4 r = ...$ (or $\log_r 4 = ...$) (One answer only is sufficient)
A1	Complete to the correct answers, **both** needed
(ii)	
M1	Combine the LHS logs to a single log. Numbers should be multiplied – if added award M0
M1	Change 1 to $\log_5 5$ and obtain a single log for the RHS
A1	Correct single log for RHS (Requires second M mark, not first)
dM1	Use LHS = RHS to obtain an equation without logs Depends on both previous M marks
A1	Correct answer
ALT 1	
M1	Combine the LHS logs to a single log. Numbers should be multiplied – if added award M0
M1	Combine the two logs on RHS
A1	Correct numerical value for **LHS**. This will need a calculator so change of base need not be seen. Equation need not be formed yet. Correct final answer implies correct value here. Otherwise min 3 sf needed
dM1	This mark requires the **first** M mark to have been given – the second M mark can be M0 Use LHS = RHS to obtain a value for $3\log_5 x$ or $\log_5 x$ Depends on both previous M marks
A1	Correct answer. This will be exact if all numbers stored on the calculator so accept 18 only.
ALT 2	
M1	Combine the LHS logs to a single log. Numbers should be multiplied – if added award M0 **Alternatively** we may see $\text{LHS} = \log_5 5 + \log_5 5832$ without ever seeing $\text{LHS} = \log_5 29160$
M1	Combine the 2 logs on RHS and change 1 to $\log_5 5$
A1	Correct RHS (Requires second M mark, not first)
dM1	Use LHS = RHS to obtain a value for x^3 Depends on both previous M marks
A1	Correct answer
ALT 3	
M1	Split $\log_5 15$ and combine all logs apart from $\log_5 5$ to a single log
M1	Combine the two logs on RHS
A1	Change $\log_5 5$ to 1 and have the correct log on LHS
dM1	This mark requires the **first** M mark to have been given – the second M mark can be M0 Use LHS = RHS to obtain a value for x^3 Depends on both previous M marks
A1	Correct answer
ALT 4	
M1	Combine the LHS logs to a single log. Numbers should be multiplied – if added award M0
M1	Combine the two logs from the RHS
A1	Obtain the equation shown
dM1	Obtain a value for x^3 Depends on both previous M marks
A1	Correct answer

Type MS · Log equations: combine into one log, then remove the log

Answer 16 of 16 · Question 4 part whole · 7 marks

2020 NOV · Paper 02 · extract 3 of 3 · find part whole in this mark scheme extract

Question Number	Scheme	Marks
	$\sum^{n}$ [illegible] $\frac{1}{}$ [illegible] $\frac{1}{}$ [illegible] *	M1M1A1cso